

Course: B. Tech Branch: Electrical Engineering and Allied Branches Semester: VI

Subject Code and Name: BTEEC602_Y22 Electrical Machine Design

Max Marks: 60

Date: 15/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level (K1-Remembering, K2- Understanding, K3 – Applying, K4 – Analyzing, K5 – Evaluating, K6 - Creating) of question or expected answer as per OBE or the course outcome (CO) on which the question is based is mentioned in front of each question.
3. The use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(CO/Level)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain the limitations considered with designing the electrical machine.	CO1/K2	6
B) Describe the nameplate of rotating machines as per standards.	CO1/K2	6
C) Discuss the insulating materials class based on thermal consideration.	CO1/K2	6
Q. 2 Solve Any Two of the following.		12
A) Illustrate a design procedure for choke (coil) design with relevant formulae.	CO2/K3	6
B) A 250 V, 37 kW, DC shunt motor has to exert a maximum torque of 150% of full load torque during the starting period. The resistance of the armature circuit is 0.2 Ohm, and the full load efficiency is 84%. The number of studs is 8. Determine ; a) The upper and lower limits of current during the starting b) The resistance of each section	CO2/K3	6
C) Show the simplex lap-wound armature with a winding table of a 4-pole generator with 24 conductors.	CO2/K3	6
Q. 3 Solve Any Two of the following.		12
A) Model an output equation for the three-phase induction motor.	CO3/K3	6
B) Compute the main dimensions of the of a 5 h.p., 400 V, 3-phase, 4-pole squirrel cage induction motor using star-delta starter. Assume the following data: Average flux density in the air gap = 0.46 Wb/m ² Ampere conductors per meter of armature periphery = 22000 Full load efficiency = 83% Full load p.f. = 0.84 lag	CO3/K3	6

Winding factor = 0.955

Stacking factor = 0.9

Current density = 4 A /mm²

No. of slots per poles per phase = 3

$L / t = 1.5$

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| C) | Illustrate the procedure of computation of losses in rotor bars and end ring of inductor machine. | CO4/K3 | 6 |
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Q. 4	Solve Any Two of the following.	12
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| A) | The thermal conductivity of assembled armature laminations is 20 times as great along the direction of laminations as in the direction across the laminations. Calculate the loss that will be conducted across the laminations in a stack 40 mm thick and 6000 mm ² in cross- section with a difference of 20°C. Given that a difference of 5°C will cause 25 W to be conducted through a cross-section of 2500 mm ² in area and 20 mm thick measured along the laminations. | CO5/K3 | 6 |
| B) | A 50 MVA turbo-alternator has a total loss of 1500 kW. Calculate the volume of air required per second and also the fan power if the temperature rise in the machine is to be limited to 30 °C. The other data given is;
Inlet temperature of air = 25 °C,
Barometric = 760 mm of mercury,
Pressure = 2 kN/m ² ,
Fan efficiency = 0.4. | CO5/K3 | 6 |
| C) | Show various types of ventilating systems in accordance with the provision of cooling ducts and how the air passes over the heated part of the machine. | CO5/K3 | 6 |

Q. 5	Solve Any Two of the following.	12
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| A) | Illustrate the types of tests performed on transformers as per the Standards for estimation of leakage reactance, resistance of winding, no load current, and losses. | CO6/K3 | 6 |
| B) | Determine winding parameters for a 100 kVA, 11 kV/415 V, 50 Hz, three-phase delta/star core type distribution transformer. Assume flux density = 1.2 T, current density = 2.5 A/mm ² , window space factor = 0.28, and constant K = 0.45 for three-phase core-type distribution transformers. Use a four-step core. | CO6/K3 | 6 |
| C) | Illustrate the step-by-step design procedure for the magnetic frame of a three-phase power transformer with relevant formulas. | CO6/K3 | 6 |

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